# Khang Nguyen | AI Engineer

Thu Duc Ward, Ho Chi Minh City – Viet Nam

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## Professional Summary

**AI Engineer** specializing in **Agentic Workflows** (LangGraph, MCP) and scalable Machine Learning pipelines. Currently pursuing a Master's degree in Data Science at HCMUS, focusing on Knowledge Representation, NLP, and Machine Learning. Proven track record in orchestrating multi-agent systems, optimizing semantic search architectures, and deploying production-ready AI solutions via **Docker** and **FastAPI**.

## Education

**University of Science (HCMUS)** — **Ho Chi Minh City**
*Master of Science in Data Science* — *2025–Present*
Focusing on **Natural Language Processing**, Knowledge Representation, and Machine Learning.

**University of Technology** — **Ho Chi Minh City**
*Bachelor in Data Science, GPA: **3.7/4.0*** — *2021–2025*
Class President; **Bronze Medalist** at Vietnam Mathematical Olympiad 2023–2024.

**AI Vietnam** — **Remote**
*AIO2023 Program* — *2023–2024*
Intensive program on advanced AI/ML & Data Science.

## Experience

**Vietnam Silicon Company** — **Ho Chi Minh City**
*AI Engineer Intern* — *May 2025–Aug 2025*
**Talent Market Intelligence & Conversational AI System**

- **System Architecture & MLOps:** Architected a highly scalable microservices platform using **FastAPI**, **RabbitMQ**, and **Docker**, reducing CV screening time by **75%** and enabling batch-processing of **100+** CVs in parallel.
- **Agentic Workflow Orchestration:** Designed a ReAct-based conversational agent utilizing **LangGraph** and **Model Context Protocol (MCP)** with **50+** integrated tools, fully automating complex HR database queries via natural language.
- **Advanced Search Pipeline:** Engineered a hybrid semantic search engine combining **pgvector (HNSW)** and **FAISS** with Sentence Transformers, achieving **92%** accuracy in skill matching with **sub-3s** pipeline latency.
- **Resilience & Scalability:** Implemented **24-API key** failover rotation for Gemini 2.5 Flash LLM, ensuring **99.5%** system uptime and continuous real-time streaming capabilities.

## Selected Projects

Multi-Agent Text2SQL System — Academic Project

**Overview:** Orchestrated an advanced Text-to-SQL architecture to bridge natural language and relational databases.

**Impact:** Developed an **autonomous schema learning** and semantic relationship discovery mechanism using **LangGraph** and multi-agent workflows, eliminating the need for manual SQL interactions.

**Tech Stack:** Python | LangGraph | LLMs | PostgreSQL

Booking.com Hotel Analytics — GitHub

**Overview:** Multi-modal prediction and segmentation pipeline for hotel reviews (text/image/metadata).

**Impact:** Fused **ResNet18** image features with tabular data for regression ($R^2 = 0.78$) and built a **stacking ensemble** for quality classification (**ROC-AUC** $= 0.89$). Packaged via **Docker** for complete reproducibility.

**Tech Stack:** PyTorch | Scikit-learn | K-Means/DBSCAN | Docker

Vietnamese Financial Sentiment Analysis — GitHub

**Overview:** Hybrid ML/DL modeling for Vietnamese stock-market headline sentiment.

**Impact:** Deployed a full NLP pipeline (**PhoBERT**, **BiLSTM**) managing data via **MongoDB**, achieving an out-of-sample accuracy of **89.2%**.

**Tech Stack:** PhoBERT XGBoost MongoDB Docker

## Technical Skills

**Programming**: Python (Advanced) SQL R JavaScript

**AI & Agentic**: LangGraph LangChain MCP ReAct Agents HuggingFace

**ML/DL**: PyTorch FAISS Scikit-learn Sentence Transformers CVXOPT

**MLOps & Infra**: Docker RabbitMQ Git API Gateway n8n Docker-Compose

**Data Engineering**: PostgreSQL pgvector MongoDB Redis Spark Kafka

## Accomplishments & Languages

**Certifications**: DeepLearning.AI – **CNN Specialization** | DataCamp – **Data Scientist Professional**

**Awards**: **2x Bronze Medals** – Vietnam Mathematical Olympiad (2023, 2024)

**Languages**: English (**IELTS 6.0** - Currently upgrading), Vietnamese (Native)